

## Sound and Waveforms

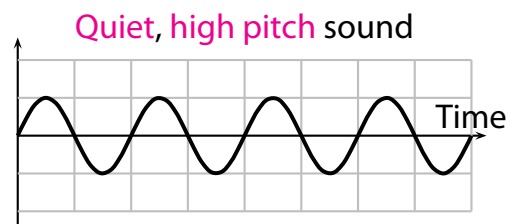
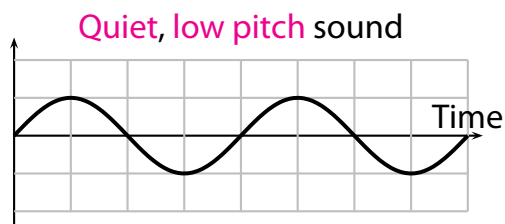
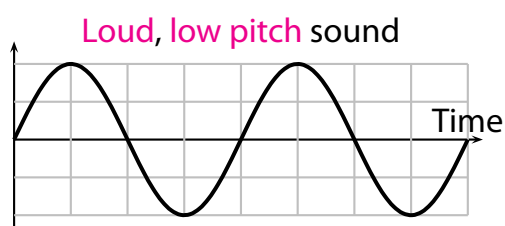
When a sound travels, the **particles oscillate** in the material it is travelling through. This means that they move forwards and backwards again and again.

The time for each oscillation back and forth is called the **time period**.

When the sound reaches our **ears** or a **microphone**, it makes a sensitive surface **oscillate** too. This can be heard as sound.

When a **microphone** is connected to a computer, a **data-logger** or an **oscilloscope**, the wave can be seen on a screen. The screen shows the **displacement** changing as **time** passes.

The greatest displacement of a wave is called its **amplitude**.



1 Look at the waves shown above and answer the questions.

(a) Describe how the loud sounds' graphs differ from the quiet sounds' graphs.

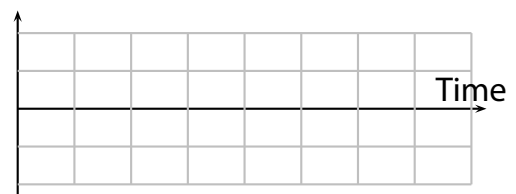
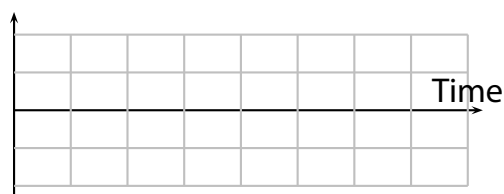
(b) Describe how the high sounds' graphs differ from the low sounds' graphs.

(c) Does a short time period mean that the sound is high pitched or low pitched?

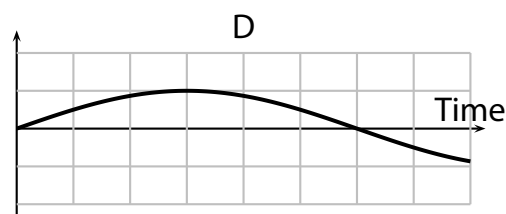
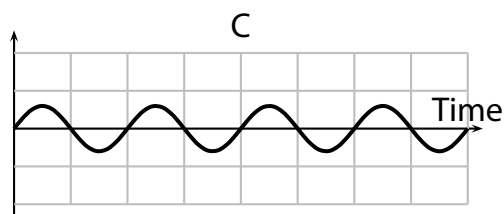
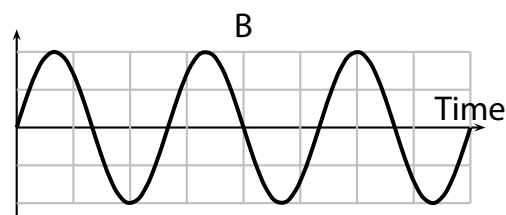
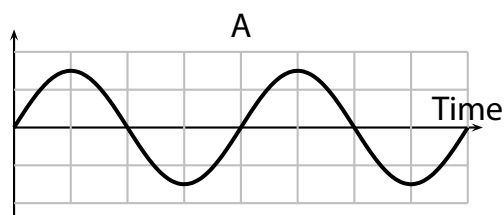
(d) Does a large amplitude mean that the sound is loud or quiet?

2 Draw waves on the graphs below to fit the descriptions.

(a) quieter and higher than the waves above (b) louder and lower than the waves above



3 Use these wave graphs to answer the questions below them.



(a) Put the letters ABCD in order to rank the waves from quietest to loudest.

(b) Put the letters ABCD in order to rank the waves from lowest to highest pitch.

(c) Which wave has the shortest time period?

The number of complete oscillations in one second is called the **frequency** of the wave.

Waves with a high frequency have a **short** time period and are **high pitched**.

4 Which wave in question 3 has the lowest frequency?

5 Complete the sentences using **20 000 Hz**, **ultrasound**, **Hz**, **20 Hz**, **hertz**.

Frequencies are measured in \_\_\_\_\_ (symbol \_\_\_\_\_).

Our ears can hear sounds with frequencies between \_\_\_\_\_ and \_\_\_\_\_.

Sounds above this range can not be heard and are called \_\_\_\_\_.

---

Each oscillation takes a very short time so the times are measured in **milliseconds (ms)**.

A millisecond is one **thousandth** of a second. There are **1000** milliseconds in one second.

We write  $1\text{ s} = 1000\text{ ms}$ , or  $1\text{ ms} = 0.001\text{ s}$ .

6 Write the times below in milliseconds.

(a) 0.002 s

(b) 0.125 s

(c) 0.02 s

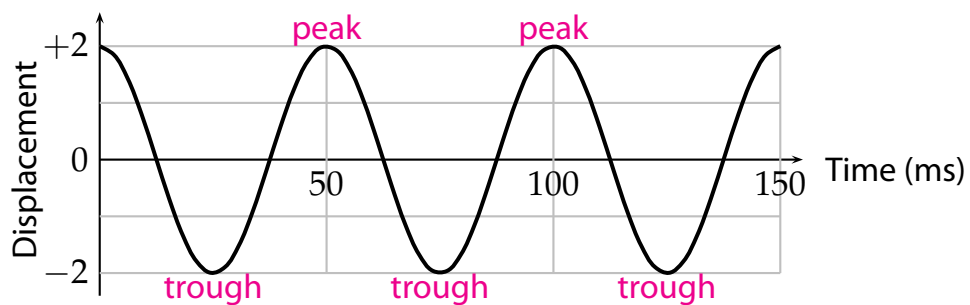
7 Write the times below in seconds.

(a) 2000 ms

(b) 3 ms

(c) 40 ms

The **time period** of this wave is 50 ms. This is the time from one **peak** to the next **peak**.



8 What is the time period of this wave in seconds?

9 In this question we look at the link between time period and frequency.

(a) A wave has a time period of 0.05 s. Work out how many times it oscillates each second using an equation.

$$\boxed{1} \div \boxed{0.05} = \boxed{\phantom{00}}$$

(b) What is the frequency of a wave if its time period is 0.05 s?

(c) A wave has a time period of 0.002 s. Work out its frequency using an equation.

$$\begin{array}{ccccc} 1 & \div & \text{time period (s)} & = & \text{frequency (Hz)} \\ \boxed{1} & \div & \boxed{0.002} & = & \boxed{\phantom{00}} \end{array}$$

(d) Work out the frequency of a wave with a time period of 0.004 s using an equation.

$$\begin{array}{ccccc} 1 & \div & \text{time period (s)} & = & \text{frequency (Hz)} \\ \boxed{1} & \div & \boxed{0.004} & = & \boxed{\phantom{00}} \end{array}$$

10 A wave has a time period of 5 ms. Calculate its frequency.

11 What is the frequency of the wave in question 8?

A frequency of 4000 Hz is usually written **4 kHz**. One kilohertz (1 kHz) is the same as **1000 Hz**.

12 Calculate the frequencies in kHz of waves with the following time periods.

(a) 10 ms

(b) 0.5 ms

(c) 40 ms

(d) 0.1 ms